Assignment: Working with HTML and JSON

This is a warm up exercise to help you to get more familiar with the HTML and JSON file formats, and using packages to read these data formats for downstream use in R data frames. In the next two class weeks, we'll be loading these file formats from the web, using web scraping and web APIs.

Select **three books** on a single subject of your choice. At least **one book must have multiple authors**. For each book, record the **title**, **authors**, and **two to three additional attributes** of your choosing (e.g., publication year, publisher, ISBN, genre, rating).

Using this information, **manually create two data files**:

- An **HTML file** containing a table with the book data (e.g., `books.html`)
- A **JSON file** representing the same data (e.g., `books.json`)

Unless you are already very comfortable with these formats, you are expected to create both files **by hand** to better understand their structure.

Next, write **R code** (using packages of your choice) to:

1. Load the HTML data into an R data frame
2. Load the JSON data into a separate R data frame
3. Compare the two data frames and determine whether they are identical

Deliverables

- `books.html`
- `books.json`
- R code that loads and compares both data sources

Package your complete solution in a **Quarto (`.qmd`) file** and **publish it to rpubs.com**.

Note: Publishing will require making your HTML and JSON files **publicly accessible on the web**.